BirdCLEF 2026

Gold Medal Solution Design

Multi-Backbone SED + Perch Ensemble with VLOM Blend,
4-Fold Cross-Validation, and 2025-Derived Techniques

BirdCLEF 2026 Team

March 2026

Abstract

We present a complete solution design for BirdCLEF 2026, a multi-label bird sound recognition competition targeting 234 species in Pantanal soundscapes. Our approach integrates three streams: (1) Google Perch TFLite models with domain-specific label heads, (2) a multi-backbone SED ensemble using EfficientNet-B0, B3-NoisyStudent, and EfficientNetV2-S trained with Asymmetric Loss and 4-fold cross-validation with model soup, and (3) inference-time techniques from 2025 top solutions. Key innovations include VLOM blending ((geometric mean + RMS)/2) for heterogeneous model fusion, domain-matched background noise augmentation with Pantanal recordings, overlapping stride inference with 1.25s TTA, and 20%/60%/20% temporal smoothing. Current public LB: **0.892**. Target: **0.926+** (competition rank #1).

Contents

1. Competition Overview

BirdCLEF 2026 is hosted on Kaggle and asks participants to identify bird species from 60-second passive-acoustic field recordings captured in the Pantanal wetlands of Brazil. The evaluation metric is **macro ROC-AUC** over 234 species.

1.1 Dataset Statistics

Split	Count	Notes
Train audio clips	35,549	Short focal recordings, variable duration
Train soundscapes	66 files	60s Pantanal field recordings, multi-label
Soundscape segments	10,658	Labeled 5-second segments
Test soundscapes	~600	Private test set
Species (classes)	234	234 primary labels
Zero-audio species	28	25 insect sonotypes + 3 amphibians (no train audio)

Table 1: Dataset overview

1.2 Key Challenges

- **Domain gap**: Train audio is focal recordings; test is passive soundscapes (ambient noise, overlapping species, varying SNR).
- **Zero-audio species**: 28 species have *no* training audio clips. Supervised only via soundscape labels.
- **Class imbalance**: Recording counts range from 1 to 500+ per species.
- **Multi-label evaluation**: Each 5-second bin can contain multiple species; macro ROC-AUC is the target.

2. Architecture Overview

Our system has three prediction streams fused at inference time: (1) **Perch stream**: Google Bird Vocalization Classifier (TFLite) with three task-specific label heads. (2) **SED stream**: Multi-backbone Sound Event Detection CNNs with attention pooling, 4-fold cross-validation, and model soup. (3) **Fusion**: VLOM blend (2025 2nd-place technique) + temporal smoothing + optional power boost.

```
Audio (60s OGG)
↓ VAD: Silero human voice removal

Perch TFLite                          SED CNN (multi-backbone)
12 clips × 3 heads (CPU)              23 stride windows + TTA (GPU)

↓                                     ↓

VLOM Blend: (geometric_mean + RMS) / 2
↓
Temporal Smooth: 20% / 60% / 20%
↓
submission.csv (234 species × 12 bins)
```

Figure 1: High-level inference pipeline

---

3. Perch Stream

We use Google's Bird Vocalization Classifier (Perch v2, CPU TFLite) as a frozen feature extractor. Perch was pre-trained on 10,000+ bird species globally and produces both a 14,795-dimensional logit vector and a 1,536-dimensional embedding. Three task-specific linear heads are trained on top of Perch outputs:

Head	Training Data	Val AUC	Weight
label_head_pseudo	Perch logits + pseudo labels	0.9748	1.0
label_head_soundscape	Perch logits + soundscape segments	0.9535	1.0
embedding_head	Perch embeddings + soundscape (nohuman)	0.9537	1.0

Table 2: Perch label head variants

---

4. SED Stream

4.1 Model Architecture

Each SED model takes a 5-second waveform (160,000 samples at 32 kHz) and produces clip-level probabilities for all 234 species.

```
Waveform (5s, 32kHz) → Mel-Spectrogram (224 × 313) ↓ EfficientNet Backbone
(pretrained, features_only) ↓ GEM Frequency Pooling (p=3, learnable) ↓
Attention SED Head (softmax attention × classification) ↓ Clip-level Sigmoid
■_clip ∈ [0,1]^234
```

Figure 2: SED model architecture

Mel-spectrogram: n_FFT=2048, hop=512, n_mels=224, f_min=0, f_max=16,000 Hz.

```
GEM(x, p) = ( (1/H) Σ x_h^p )^(1/p), p initialized at 3.0, learned during training
```

$$\blacksquare\_clip = \sigma\left(\Sigma_t \alpha_t \cdot \phi_t \right), \quad \alpha_t = \mathrm{softmax}(\tanh(W_a \cdot f_t))$$

4.2 Backbone Strategy

Backbone	Pretraining	Params	Ens. Weight	Notes
tf_efficientnet_b0.ns_jft_in1k	NoisyStudent JFT	5.3M	1.0	Baseline anchor
tf_efficientnet_b3.ns_jft_in1k	NoisyStudent JFT	12.2M	2.0	2025 dominant backbone
tf_efficientnetv2_s.in21k_ft	ImageNet-21k	21.5M	2.0	2025 2nd/5th place

Table 3: Backbone comparison

B3-NoisyStudent (tf_efficientnet_b3.ns_jft_in1k) was used by multiple top-10 solutions in BirdCLEF 2025. NoisyStudent pretraining on JFT-300M provides significantly stronger acoustic feature representations than plain ImageNet initialization.

4.3 Training Recipe

Experiment	Key Change	Holdout AUC
sed-b0-v5 (BCE)	Baseline: BCE, clip-only loss	0.9192
sed-b0-v6	Fold-only val (no soundscape)	0.7543
sed-v2s-v1	EfficientNetV2-S backbone	0.7980
sed-b0-v9-asl	ASL (γ■=4, γ■=0, clip=0.05)	0.9467 ★
sed-b0-v10-cutmix	ASL + CutMix	0.9391
sed-b0-v11	ASL + soft secondary labels	0.9359
sed-b0-v12-bce	BCE with dual loss (repeat)	0.9079
sed-b0-v15-no-sec	ASL, no secondary labels	0.9176

Table 4: Ablation results — holdout AUC on held-out soundscapes. ★ = best single model.

Key finding: ASL delivers +0.027 holdout AUC over BCE — the single most impactful technique.

4.4 Asymmetric Loss

Asymmetric Loss (ASL) addresses class imbalance in multi-label classification by applying different focusing parameters to positives and negatives:

$$L\_ASL = \{ (1-p)^{\wedge}\gamma+ \cdot \log(p)\ [\text{positive}]\ \{ (p - m)+^{\wedge}\gamma- \cdot \log(1-p+m)\ [\text{negative}]\ \gamma+ = 0 \text{ (no positive focusing) } \gamma- = 4 \text{ (strong negative down-weighting) } m = 0.05 \text{ (probability margin clip)}$$

ASL effectively ignores easy negatives while maintaining strong gradient on positives, critical for the highly imbalanced multi-species detection task.

4.5 Background Noise Augmentation

A key technique from 2025 top solutions is background noise injection using Pantanal soundscapes as the noise corpus:

$$x\_aug = x\_clean + \alpha \cdot x\_noise, \quad SNR \sim U(5, 30)\ dB$$

We use birdclef-2026/train_soundscapes/*.ogg (10,658 files) as the noise corpus. These are real Pantanal field recordings — the *same domain* as the test set — providing ideal domain adaptation with zero additional data download required.

5. 4-Fold Cross-Validation + Model Soup

A single-model SED achieves holdout AUC ≈ 0.947. 4-fold CV with model soup yields approximately +0.005–0.012 via: (1) using all training data across folds, (2) geometric-mean ensemble of 4 fold predictions (variance reduction), and (3) model soup within each fold (top-3 checkpoints weight-averaged).

Fold	Train clips	Val clips	Val soundscapes
0	26,662	8,887	16 files
1	26,661	8,888	17 files
2	26,662	8,887	16 files
3	26,662	8,887	17 files

Table 5: Stratified 4-fold split statistics

$$\theta_{soup} = (1/k)\ \Sigma\ \theta_i\ \text{[model soup: top-k weight average]}\quad p_{fold\text{-}ensemble} = \exp(\ (1/K)\ \Sigma\ \ln p_k\ )\ \text{[geometric mean across folds]}$$

Geometric mean is preferred over arithmetic mean for probabilities as it handles heterogeneous confidence scales without over-confident blending.

6. Pseudo-Label Pipeline

The 66 training soundscapes have many unlabeled 5-second segments. We expand supervision using Perch-generated pseudo labels across 5 iterative rounds:

- **Round 1**: Perch ensemble on all 10,658 soundscape segments. Filter top confidence predictions per species.
- **Rounds 2–5**: Train SED with round-n pseudo labels, generate round-(n+1) from trained SED + Perch.
- **Hard pseudo**: Per-clip binary labels for soundscape data augmentation (extra_soundscape_csv).
- **Soft pseudo**: Continuous scores for knowledge distillation (soft_pseudo_csv, 10× oversample).

Current: Round 5 pseudo labels (outputs/pseudo_soundscape_labels_r5.csv), providing 1,168 additional labeled segments beyond the original 312 soundscape annotations.

7. Inference Pipeline

7.1 Human Voice Removal (VAD)

Silero VAD detects human speech and truncates audio before speech begins (if speech duration ≥ 2s and onset ≥ 8s). This removes announcer segments common in competition recordings.

7.2 Overlapping Stride Windows

We use STRIDE=2.5s sliding windows instead of standard non-overlapping 12-clip inference:

- Window size: 5s (160,000 samples at 32 kHz)
- Stride: 2.5s (50% overlap)

- Windows per 60s file: 23 (vs 12 non-overlapping)
- Output bins: 12 (each bin averages 2–3 overlapping windows)

Overlapping windows provide ~2× more inference samples with context overlap, significantly improving detection of species present at window boundaries.

7.3 Test-Time Augmentation (TTA)

For each set of 23 windows, we run inference on a 1.25s circular shift:

```
■_TTA = 0.5 · ■_original + 0.5 · ■_shifted
```

This provides phase-invariant predictions at the cost of one additional forward pass per model.

---

8. Post-Processing

8.1 VLOM Blend (2025 2nd Place)

Simple weighted averaging underperforms for heterogeneous models with different confidence scales. We use the VLOM blend from the 2025 2nd-place solution — the average of weighted geometric mean and weighted RMS:

```
■_VLOM = ( exp(w_P · ln ■_P + w_S · ln ■_S) + sqrt(w_P · ■_P² + w_S ·
■_S²) ) / 2 w_P = 0.55 (Perch weight), w_S = 0.45 (SED weight)
```

The geometric mean captures relative confidence ordering; RMS amplifies strong signals. Their average handles both high-confidence agreement and low-confidence disagreement.

8.2 Temporal Smoothing 20%/60%/20% (Universal 2025)

```
■'_i = 0.20 · ■_{i-1} + 0.60 · ■_i + 0.20 · ■_{i+1}
```

Boundary: edge clips add the missing neighbor weight to self (clip 0: 80% self + 20% next; clip 11: 20% prev + 80% self). This weighting was used universally across 2025 BirdCLEF top-10 solutions.

8.3 File-Max Trick (TRICK1)

```
■'_i = ■_i + 0.05 · max_j ■_j
```

Species present strongly at any point receive a baseline lift across all clips, exploiting Pantanal habitat characteristics where species tend to persist.

8.4 Power Boost (2025 3rd Place — Conditional)

```
■'_{i,c} = { ■_{i,c}^γ+ if c ∈ top-N species per clip γ+ = 0.5 (boost) {
■_{i,c}^γ- otherwise γ- = 1.5 (suppress) N = 3, disabled by default
```

Status: DISABLED by default. Enable only after confirming improvement on *both* validation and holdout AUC. Never accept based on public LB alone.

---

9. Experiment Schedule and Expected Results

Phase	Experiment	GPU	Key Contribution
Phase 2	v21-dual-rating3	GPU0	Dual loss + rating filter
Phase 2	v22-dual-noclipmix	GPU0	No CutMix in dual-loss

Phase	Experiment	GPU	Key Contribution
Phase 2	V2S-v2-asl	GPU0	EfficientNetV2-S + ASL
Phase 2	v28-final-combo	GPU0	ASL + dual + soft-KD + circ_shift
Phase 2	v24-soft-pseudo	GPU1	Soft pseudo knowledge distillation
Phase 2	v26-asl-npcen	GPU1	ASL, no PCEN (ablation)
Phase 2	v27-soft-boost	GPU1	Soft pseudo boost weight
Phase 3	v30-bgnoise	GPU1	Background noise augmentation
Phase 3	B3-v1-asl	GPU0	B3-NS backbone, full recipe
Phase 3	B3-fold0..3	both	4-fold B3 geometric-mean ensemble
Phase 4	V2S-fold0..3	TBD	4-fold V2S (pending B3 comparison)

Table 6: Experiment schedule

Milestone	Expected LB	Delta vs 0.892
Current (v9-asl-soup + Perch 3-head)	0.892	baseline
Phase 2 best single model (v28)	0.895–0.903	+0.003–+0.011
V2S-v2-asl single model	0.900–0.915	+0.008–+0.023
B0 4-fold soup ensemble	0.905–0.920	+0.013–+0.028
B3 4-fold soup ensemble	0.912–0.928	+0.020–+0.036
Multi-backbone (B0+B3+V2S) + PP	0.915–0.933	+0.023–+0.041

Table 7: Expected LB progression. Target: 0.926 (rank #1 as of 2026-03-18)

10. Key Decisions and Trade-offs

- **B3-NS over EfficientNetV2-B3**: tf_efficientnet_b3.ns_jft_in1k (NoisyStudent JFT-300M, 12.2M params) matches the exact variant used by 2025 top-10 solutions. V2-B3 (14.4M) has only ImageNet-21k pretraining.
- **Domain-matched noise**: Using competition train soundscapes as noise corpus (vs ESC-50/AudioSet) ensures Pantanal-specific characteristics match test environment.
- **VLOM over weighted average**: Simple weighted avg is suboptimal when Perch (calibrated sigmoid) and SED (raw sigmoid) have different confidence scales.
- **3-tap over 5-tap smooth**: Previous 5-tap [0.05,0.15,0.60,0.15,0.05] with sharpening adds complexity without consistent validated gain. Plain 3-tap [0.20,0.60,0.20] matches 2025 universal standard.
- **Competitor models excluded**: All SED checkpoints are our own trained models only.
- **Power boost disabled by default**: PP that conditions on test predictions can produce false public LB improvements that do not generalize. Validate on holdout AUC first.

11. Validation Framework

A fixed holdout set of 13–17 soundscape files per fold is kept separate throughout all experiment development. **Holdout AUC is the ground truth**; validation AUC (soundscape segments only) is used for early stopping and checkpoint selection.

```
Submit if: holdout_AUC > 0.9193 (v5-BCE baseline benchmark)
```

Current best: sed-b0-v9-asl-soup at holdout AUC = **0.9467** (Δ = +0.0274 over baseline).

Post-processing validation rule: Each PP step must improve *both* validation and holdout AUC. Never accept improvements based solely on public LB.

12. Techniques from 2025 Top Solutions

Technique	Source	Status	Est. Gain
B3-NoisyStudent backbone	Multiple 2025 top-10	Phase 3	+0.005–0.018
Background noise augmentation	2025 3rd–5th place	Phase 3 (v30)	+0.003–0.008
ASL loss	2025 top solutions	Done (v9)	+0.027 ✓
4-fold CV + model soup	Standard 2025 practice	Phase 3	+0.005–0.012
VLOM blend (geomean+RMS)/2	2025 2nd place	In notebook	+0.002–0.005
Temporal smooth 20/60/20	Universal 2025	In notebook	+0.001–0.003
Overlapping stride windows	2025 top solutions	Done (v6-TTA)	+0.002–0.005
TTA circular shift	2025 top solutions	Done (v6-TTA)	+0.001–0.003
Soft pseudo KD	2025 top solutions	Phase 2 (v24)	+0.002–0.006
Power boost	2025 3rd place	Conditional	+0.001–0.003

Table 8: 2025 top-solution techniques — status and estimated contribution

13. Conclusion

We present a comprehensive gold-medal targeting solution for BirdCLEF 2026 that addresses the core challenges:

- **Domain gap**: Background noise augmentation with Pantanal recordings, soundscape oversampling.
- **Zero-audio species**: Perch 3-head ensemble, 5-round pseudo label pipeline.
- **Class imbalance**: Asymmetric Loss (γ■=4), soft pseudo knowledge distillation.
- **Heterogeneous model fusion**: VLOM blend, multi-backbone geometric mean.
- **Temporal structure**: Overlapping stride windows, TTA, 20%/60%/20% smoothing.

The designed system builds incrementally on a proven foundation (current LB=0.892, holdout=0.9467) and targets LB 0.926 through systematic application of 2025-validated techniques. Each component has been validated through ablation experiments or verified from 2025 top-solution writeups, ensuring expected gains are grounded in evidence.

. References

[1] Ridnik, T. et al. (2021). Asymmetric Loss For Multi-Label Classification. ICCV 2021.

[2] Wortsman, M. et al. (2022). Model soups: averaging weights of multiple fine-tuned models. ICML 2022.

[3] Tan, M. & Le, Q.V. (2021). EfficientNetV2: Smaller Models and Faster Training. ICML 2021.

[4] BirdCLEF 2026 Competition. (2026). Identify bird species in field recordings. Kaggle.

[5] Kahl, S. et al. (2021). BirdNET: A deep learning solution for avian diversity monitoring. ECOLIND.